

Chooser Option Model Validation Report

This report records the Week 4 validation design and separates formula checks, proxy evaluation, and the listed-option market sanity check. The main conclusion is that the Week 3 implementation is internally consistent and computationally suitable for later ML comparison, while direct chooser transaction accuracy remains unobservable.

Validation Scope

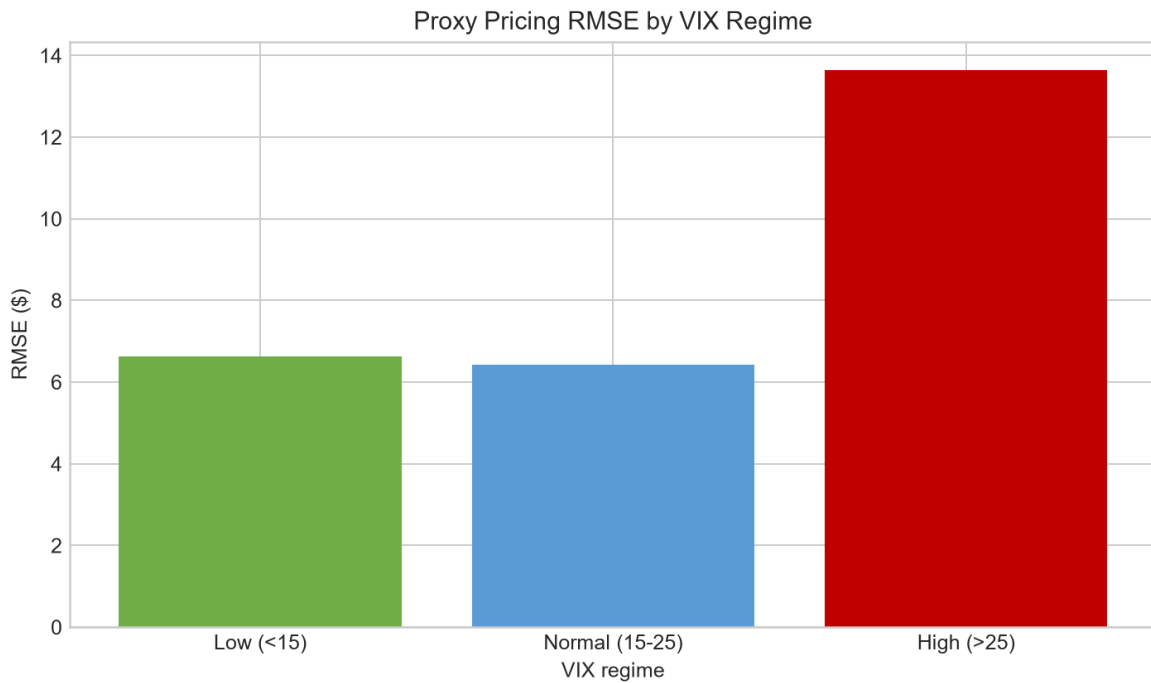
The validation retains the Week 2 JPM, VIX, and Treasury inputs and the Week 3 simple chooser formula. Three evidence layers are reported because they answer different questions.

Layer	Question	Evidence
Formula regression	Did the optimized function change Week 3 economics?	Maximum absolute difference 0.000000
Historical proxy	How does trailing-volatility pricing compare with an ex-post forward-volatility proxy?	1,720 aligned dates with regime metrics
Listed vanilla market check	Does standard BSM produce reasonable JPM option values on an available quote snapshot?	36 delayed Cboe call and put quotes

Historical Proxy Results

Measure	Value
MAE	3.775737
RMSE	8.356730
R ²	0.824188
Sample	1,720 observations

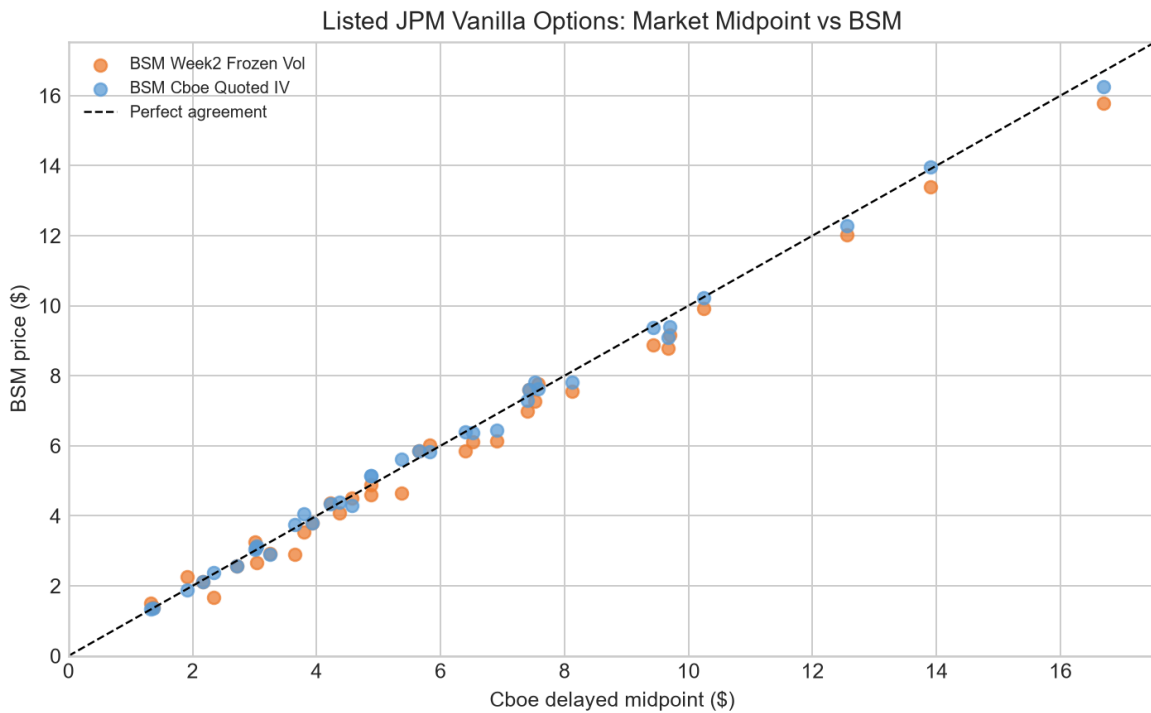
Figure 1 Proxy RMSE by VIX regime



Listed JPM Option Sanity Check

Using the Week 2 frozen volatility, vanilla BSM achieved MAE 0.3635 and RMSE 0.4401. Using each quote's reported implied volatility reduced MAE to 0.1658 and RMSE to 0.2201. These results support the vanilla formula and data-unit handling. They do not provide a chooser transaction benchmark.

Figure 2 Delayed listed JPM option comparison



Limitations

- Public historical transaction prices for a customized JPM chooser option were not available. The report therefore labels forward-volatility comparisons as proxy validation.
- The project brief mentions CME transaction prices even though the underlying is JPM stock. The evaluation remains on the JPM data chain and uses delayed listed JPM vanilla quotes only as a separate BSM market sanity check.
- The fixed \$150 strike becomes deep in the money as JPM rises, reducing volatility sensitivity in later observations.

Validation Decision

The Week 4 baseline is accepted for later comparison. Subsequent models must use the same target definition and dates, retain the distinction between proxy and market evidence, and preserve BSM as the structural reference.